Reading between the letters. Exploring biases, gaps, and context in historical correspondence data

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Introduction

Working with historical data in combination with digital methods is a messy and often unsatisfactory matter: Historical sources are incomplete, ambiguous, and biased, and so is the data extracted from these sources. Moreover, specific forms of historical sources come with additional challenges: Data based on historical letters bear an even bigger risk regarding gaps and biases due to the letter's event-based character (Fischer 2023). Historical letters, as part of a correspondence, were distributed over several locations and, as a piece of literature, often forwarded and passed around between various historical agents. Therefore, the estates of letters are the result of an awareness of the importance of their collection, be it of the correspondents themselves or their descendants. This has several consequences: Letters were often destroyed when considered too personal, lost due to historical circumstances, or collection efforts focused on specific individuals, mirroring the *zeitgeist* of the respective era. On top of these challenges, efforts of (retro-)digitization have been selective as well, in many instances reproducing and reinforcing past practices of knowledge production and being heavily skewed towards certain social groups and geographic regions (Zaagsma 2022).

On the other hand, the data and metadata of letters contains a multitude of implicit knowledge and relations. When working with quantitative methods, letter corpora are a great example for how a change in perspective enables us to find previously unconsidered patterns. Letters have therefore been the focus of numerous studies in digital humanities, especially in the field of historical network research, due to their suitability for exploring questions of communication structures and practices (Ahnert / Ahnert 2015; Ahnert / Ahnert 2019; Błoch et al. 2020; McShane 2018) , or in tracking the emergence of fields of research (Rossini 2024).

Against this background, the following question arises: How can we balance new insights on historical subject matters with a critical reflection of the historicity of the sources, their gaps and biases, and ultimately prevent circular reasoning fallacies? And how can we go further, using data-driven methods, such as Semantic Web technologies, to create more complex, diverse perspectives based on our existing data?

This submission aims to explore these questions on multiple levels. Using a dataset of correspondences of actors from the period of Early Romanticism, a European literary period, we aim to uncover gaps and biases in the data, investigate their impact, and employ a dual strategy of mitigating these gaps through LOD-approaches and incorporating their effects into our analyses. For this purpose, we are developing a mixed methods-approach including the use of knowledge graphs, data visualisation, and historical network research.

Use Case

Our corpus is based on correspondences of the “Romantic group” and their correspondents from 1790 to 1802. The Romantics were a group of young German authors and scholars, mainly based in Jena (and Berlin), who revolutionized the literary scene at the time through their collaborative working methods, their new perspective on novels, and their numerous translations, journal articles and reviews. Letters played an important part for the Romantics, as they had a group-forming, stabilizing and identity-forming function (Bamberg 2020). The corpus represents part of the *Bildungsbürgertum* or intelligentsia of the turn between the 18th and 19th century, including female and male authors, scholars and publisher of different ages, educational levels, and religious groups.

Our research is part of the ongoing DFG-Project “Correspondence of early Romanticism. Edition – Annotation – Network Research” which aims to collect and publish around 7,000 of these letters to gain insights into processes of epistolary communication and knowledge transfer. The corpus is based on both manuscripts and printed editions. It also includes inferred letters, i.e. letters

which no longer exist as a physical source, but for which we have evidence in existing letters. In the process, we collect metadata, and full texts of the letters and annotate mentions of persons, institutions, works, periodicals, bibliographic references, and places as well as semantic statements in form of SPO-Triples (Subject – Predicate – Object) concerning knowledge exchange and communication processes. These data are transformed into a knowledge graph (Neo4J) which is based on a project-specific ontology (Deicke et al. 2024). From this graph, we abstract datasets of nodes and edges to analyse within a framework of historical network research and to represent in network visualizations.

In theory, the metadata in combination with the content of epistolary communications opens up a wide field of information. Yet in practice, the data is highly biased and provides a limited perspective. These challenges are intensified through the collection of letters from printed editions: Past editorial decisions are mirrored in the data, e.g. modern deletions in historical letter texts (Wernli 2022) . Letters of women as a “minoritized” group, as persons who are in opposition to a more powerful social group (D’Ignazio / Klein 2020) , were often only regarded relevant when exchanged with men (Becker-Cantarino 2000). Additionally, numerous letters are missing from the sequence of correspondences and hence, in the network. On a more basic level, editions of letters are generally centred on the correspondence activity of one specific person, focusing only on the letters written and received by this author or scholar. Consequently, their influence on the data potentially eclipses other perspectives, and their weight in the network skews most quantitative measures.

Methods

Enriching data

One way to balance this missing contextual knowledge is to add information from other data sources to the knowledge graph. In doing so, we also address the situated knowledge reflected in the sources (and the data), acknowledging that knowledge is produced by specific people in specific circumstances (Haraway 1988). In our case, this affects a) the historicity of the source, b) the *zeitgeist* of the reused editions, and c) our own editorial and conceptual decisions.

We add data from the German National Library’s Integrated Authority File (GND) and Wikidata. In preparation, we select which available information is relevant for our research questions and critically reflect on the quality and biases of this data. As a result, we can supplement our database with information about the gender and familiar relationships of individuals, facilitating a more comprehensive picture of our knowledge domain and expanding the potential for analysis.

Reasoning

Based on the developed ontology, we are further able to make implicit information within the letters explicit and machine-readable. This concerns basic graph completion but also advanced inferences. For example, through inferring co-location, we focus on channels of communication beyond the written word, i.e., personal encounters in salons, universities, and private gatherings. When persons who know each other (proven through exchanged letters) are at the same time in the same place, we can assume with a certain probability that they encountered one another. Reasoning routines in the knowledge graph make these and other implicit knowledge accessible for computational analysis, enhancing our understanding of the communication network.

Visualization

However, despite the measures described above, some biases and gaps remain in the data. Therefore, we add another layer of visual exploration to the methods with which we address the limitations of our data. Following Hengesbach et al. (2020), we ask how we can “see what is not shown” in order to make the specific perspective of our dataset visible (Ahnert et al. 2020). In particular, this concerns the fundamental difference between members of the “Romantic group” and those persons who are only included in the network because of their correspondence with one of these members. This difference has far-reaching consequences for analysis and interpretation, yet it routinely disappears in visualisations that treat each node equally. In doing so, we ground our work in the positionings of authors such as Jessop (2008), Drucker (2011) and Kleymann and Stange (2021), and employ methods of data visualisation as a hermeneutic tool to contextualize our analyses and interpretations.

Conclusion

Through a mixed-methods approach of combining information from different databases, semantic reasoning, and visualization, we are crea-

ting a diversified and multilayered knowledge base and providing a critical lens for exploring and analysing historical correspondence data.

However, a critical validation of these methods is crucial to ensure data quality and reproducibility of results. As Johanna Drucker puts it: "Computers do not interpret; they simply find patterns." (Drucker 2017) Modelling means defining the ways for finding these patterns, yet incorporating data from multiple sources also means incorporating multiple models. Hence, the conceptual choices behind these models are reflected in the data as well (e.g., the choice to model gender as a binary concept). Additionally, enrichment of a knowledge base via norm data may even amplify biases because minoritized groups often lack representation in authority records.

To not add a layer of "retro-humanism" to our data (Risam 2015), these decisions must be well-considered, documented and contextualized. In our data-critical approaches, we must avoid binary categories – certain or uncertain, true or false – and instead remember that "history is about the production of evidence-based arguments, not about telling the truth." (Fickers 2021) So instead of trying to fit historical sources to a binary system, we focus on finding best practices to produce "evidence-based arguments" to interpret the results we gain from these always incomplete, ambiguous and ambivalent historical records.

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